

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q20-1535

Skaggs Island 138 kV
June 25, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1535. The Interconnection Customer proposes a 78.3 MW solar photovoltaic facility interconnecting at the Skaggs Island 138 kV substation in the Virginia area. The request was received on July 28, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 408 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$332,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Northern Neck 345/138kV Auto
Contingency: Skaggs Island Jct 2 - Chancellor Row Substation 345 kV
Loading: 111.3% (Rating: 496 MVA)
DFAX: 6.1%
Pre-project loading: 110.4%
2. Facility: Yellow Crk - N Randolph Substation 138 kV
Contingency: Big Hanaford - Fritch 138 kV
Loading: 100.6% (Rating: 241 MVA)
DFAX: 6.0%
Pre-project loading: 98.6%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Northern Neck 345/138kV Auto	Replace transformer with higher-rated unit	\$13,901,000	2% (\$273,000)
Yellow Crk - N Randolph Substation 138 kV	Reconductor line	\$4,113,000	1% (\$59,000)
	Total allocated to project		\$332,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.